



BUILDCORED

ORCAS v1.5

30 Days. 30 Projects. Zero Hardware Required.

30

PROJECTS

4

WEEKS

~1h

PER DAY

0

CLOUD DEPS

Build one project every day for 30 days. Each project teaches hardware engineering thinking — using only your laptop, webcam, and Python.

Mac, Windows, and Linux. All AI runs locally.

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WEEK 1

BODY AS INPUT

Your body is the controller

01	RockLook	Look down and rock music starts playing. Look up and it pauses.	Easy
02	AirCanvas	Pinch your fingers to draw on your screen. Like a magic pen.	Easy
03	VolumeKnuckle	Raise your fist to turn volume up. Lower it to turn it down.	Easy
04	BlinkLock	Blink three times fast to lock your screen. Wink to unlock.	Moderate
05	FaceEQ	Turn your head to scrub a song. Tilt to change the pitch.	Moderate
06	BreathClock	Your computer listens to your breathing and counts BPM.	Moderate
07	KeyboardOscilloscope	Press keys to create tones. Multiple keys form visible chords.	Easy

WEEK 2

LOCAL AI CORE

AI that runs on your laptop

08	PocketAgent	Run an AI assistant entirely on your laptop. No internet.	Moderate
09	WhisperDesk	Talk to your laptop and it types what you say. Offline.	Moderate
10	TerminalBrain	Terminal watches for errors, suggests fixes using local AI.	Advanced
11	MoodSynth	Type a mood and AI creates ambient sounds to match.	Moderate
12	SnapAnnotator	Webcam snaps a photo, AI labels everything it sees.	Moderate
13	DailyDebrief	AI reads your git commits and writes a summary of your day.	Moderate
14	RegisterBot	Build a tiny computer inside your computer. The whole thing.	Advanced

WEEK 3

SIGNALS & SYSTEMS

The logic behind real hardware

15	AudioScope	See sound. Frequencies dance on screen from your mic input.	Moderate
16	EchoKiller	Record your voice with echo, then build a filter to remove it.	Advanced
17	PWMSimulator	Slide a control, watch a square wave change. A virtual LED dims.	Easy
18	DepthMapper	Webcam estimates distance to everything. Paints a depth map.	Advanced
19	I2CPlayground	Simulate how chips talk to each other. Watch signals animate.	Advanced
20	SensorLogger	Laptop becomes a live sensor dashboard: mic, camera, CPU.	Moderate
21	UDPOscilloscope	Send sensor data over your network, display as oscilloscope.	Moderate

WEEK 4

FULL SYSTEMS

Combine everything. Ship it.

22	CircuitWhisperer	Draw a circuit on paper, photograph it, AI reads it.	Moderate
23	ObjectFollower	Point camera at a colored object, your code tracks it live.	Moderate
24	HardwareTA	AI reads hardware datasheets and answers your questions.	Advanced
25	FirmwarePatcher	Load a firmware file, AI finds hidden patterns in raw bytes.	Advanced
26	CacheSim	Simulate CPU memory cache. Watch hits, misses, evictions.	Advanced
27	SignalFlags	Teach your computer to read semaphore signals from your hands.	Moderate
28	MorseDecoder	Tap spacebar in Morse code, computer decodes the message.	Moderate
29	SilentAssistant	Camera sees you, AI understands, responds, speaks back.	Expert
30	OrcaOS	Combine the best of 30 days into one personal OS. Ship it.	Expert